



Steps

Design Req.



Op. Points



Magnetic Builder



Summary

Tool menu

Actions



DATASHEET

OpenMagnetics

JULY 6, 2026

My custom magnetic

EQ 13/3 67

Power Transformer



CORE LOSS

37.89 μ W



WINDING LOSS

14.70 μ W



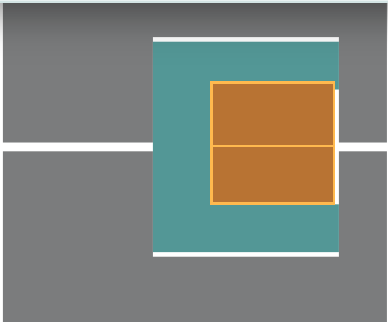
PEAK B

214.52 μ T



CORE

EQ 13/3



Show H Field

Show E Field

⚡ ELECTRICAL SPECIFICATIONS

PARAMETER	SYMBOL	TYPICAL	UNIT	CONDITIONS
Operating Frequency	f	8	MHz	
Peak Flux Density	B_{pk}	214.52	μ T	At operating point

Primary $f = 8.0 \text{ MHz}$ **Voltage***sinusoidal*

V_{PK-PK}	V_{PK}	V_{RMS}	V_{OFFSET}	DUTY
0.87 V	0.43 V	0.3 V	64.23 μV	50.0%

**Current***sinusoidal*

I_{PK-PK}	I_{PK}	I_{RMS}	I_{OFFSET}	DUTY
0.12 A	57.97 mA	40.5 mA	-466.83 μA	50.0%

Secondary $f = 8.0 \text{ MHz}$ **Voltage***sinusoidal*

V_{PK-PK}	V_{PK}	V_{RMS}	V_{OFFSET}	DUTY
43.32 mV	21.66 mV	15.32 mV	3.21 μV	50.0%

**Current***sinusoidal*

I_{PK-PK}	I_{PK}	I_{RMS}	I_{OFFSET}	DUTY
2.3 A	1.16 A	0.81 A	-9.34 mA	50.0%





Core Material	67
Number of Stacks	1
Effective Length (le)	18.24 mm
Effective Area (Ae)	20.03 mm ²
Effective Volume (Ve)	365.36 mm ³
Minimum Area	19.14 mm ²
Total Gap Length	450 μm
Number of Gaps	3

↔ CORE GAPPING

#	TYPE	LENGTH
1	Additive	150 μm
2	Additive	150 μm
3	Additive	150 μm

⚡ MATERIAL PROPERTIES

Manufacturer	Fair-Rite
Curie Temperature	450 °C
Density	4.8 Mg/m ³

📦 WINDING CONFIGURATION

WINDING	TURNS	PARALLELS	WIRE
Primary	2	1	Rect. 2.00x1.00 mm
Secondary	1	1	Dummy

📊 PERFORMANCE ANALYSIS

Test Conditions: $f = 8.0 \text{ MHz}$, $T_{amb} = 25^\circ\text{C}$

Core Losses (Pcore)	37.894 μW
Ohmic Losses	887.053 nW
Skin Effect Losses	13.806 μW
Proximity Losses	2.359 nW
Total Losses	52.59 μW

📈 WINDING PERFORMANCE

WINDING	DCR	LOSS	LEAKAGE
Primary	540.64 μΩ	14.70 μW	-
Secondary	-	0 W	-



ⓘ Auto-generated from design specs and simulation. Verify before production use.

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